ATOMS

1. In a Rutherford α scattering experiment when a projectile of charge Z_1 and mass M_1 approaches a target nucleus of charge Z_2 and mass M_2, the distance of closest approach is r_0. The energy of the projectile is:

(1) directly proportional to $M_1 \times M_2$

(2) directly proportional to Z_1Z_2

(3) directly proportional to Z_1

(4) directly proportional to mass M_1

2. The Bohr model of Atom:

(1) Assumes that the angular momentum of electrons is quantised.

(2) Uses Einstein's photoelectric equation.

(3) Predicts continous emission spectra for atoms

(4) Predicts the same emission spectra for all types of atoms.

3. According to Bohr's theory of hydrogen atom, for the electron in the n^{th} allowed orbit the:

I. Linear momentum is proportional to 1/n.

II. Radius is proportional to n.

III. Kinetic energy is proportional to $1/n^2$.

IV. Angular momentum is proportional to n.

Choose the correct option from the codes given below:

(1) I, III and IV

(2) Only I

(3) I and II

(4) Only III

4. A hydrogen atom and Li^{++} ion are both in the second excited state. If L_H and L_{Li} are their respective electronic angular momenta and E_H and E_{Li} their respective energies, then:

(1) $L_H > L_{Li}$ and $|E_H| > |E_{Li}|$

(2) $L_H = L_{Li}$ and $|E_H| < |E_{Li}|$

(3) $L_H = L_{Li}$ and $|E_H| > |E_{Li}|$

(4) $L_H < L_{Li}$ and $|E_H| < |E_{Li}|$

5. The minimum energy to ionise an atom is the energy required to:

(1) add one electron to the gaseous state of atom

(2) excite the atom from its ground state to its first excited state.

(3) remove one outermost electron from the gaseous state of atom

(4) remove an innermost electron from the gaseous state of atom

6. As an electron makes transition from an excited state to the ground state of a hydrogen like atom/ion:

(1) Kinetic energy decrease, potential energy increases but total energy remains same

(2) Kinetic energy and total energy decrease but potential energy increases

(3) Its kinetic energy increases but potential energy and total energy decreases

(4) Kinetic energy, potential energy and total energy decrease

7. If we have to apply Bohr model to a particle of mass m and charge q moving in a plane under the influence of a magnetic field B, the energy of the charged particle in the n^{th} level will be:

(1) $n\left[\frac{hqB}{2\pi m}\right]$

(2) $n\left[\frac{hqB}{4\pi m}\right]$

(3) $n\left[\frac{hqB}{8\pi m}\right]$

(4) $n\left[\frac{hqB}{\pi m}\right]$

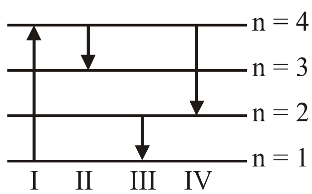
8. In a hypothetical system, a particle of mass m and charge (–3q) is moving around a very heavy particle charge q. Assume that Bohr's model is applicable to this system, then velocity of mass m in first orbit is:

(1) $\frac{3q^2}{2\varepsilon_0 h}$ (2) $\frac{3q^2}{4\varepsilon_0 h}$

(3) $\frac{3q}{2\pi\varepsilon_0 h}$ (4) $\frac{3q}{4\pi\varepsilon_0 h}$

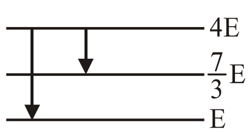
9. An electron of energy 11.2 eV undergoes an inelastic collision with a hydrogen atom in its ground state. [Neglet recoiling of atom as $m_H >> m_e$]. Then in this case:
(1) The outgoing electron has energy 11.2 eV
(2) The entire energy is absorbed by the H-atom and the electron stop.
(3) 10.2 eV of the incident electron energy is absorbed by the H-atom and electron would come out with 1.0 eV energy.
(4) None of the above

10. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



(1) III (2) IV (3) I (4) II

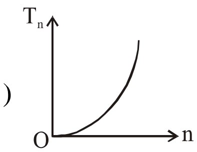
11. The following diagram indicates the energy levels of a certain atom when the system moves from 4E level to E. A photon of wavelength λ_1 is emitted. The wavelength of photon during its transition from $\frac{7}{3}$E level to E is λ_2. The ratio $\frac{\lambda_1}{\lambda_2}$ will be :

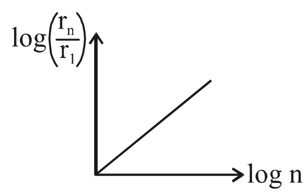


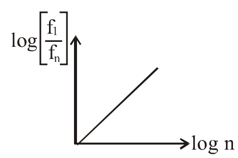
(1) $\frac{9}{4}$ (2) $\frac{4}{9}$ (3) $\frac{3}{2}$ (4) $\frac{7}{3}$

12. The wavelengths involved in the spectrum of deuterium $\left({}^2_1D\right)$ are slightly different from that of hydrogen spectrum, because:
(1) Size of the two nuclei are different
(2) nuclear forces are different in the two cases
(3) masses of the two nuclei are different
(4) Attraction between the electron and the nucleus is different in the two cases.

13. If in hydrogen atom, radius of n^{th} Bohr orbit is r_n. Time period and frequency of revolution of electron in n^{th} orbit are T_n and f_n choose the correct option:

(1) 

(2) 

(3) 

(4) All of the above

14. The hydrogen ion consists of:
(1) One neutron only
(2) One proton only
(3) One proton and one electron
(4) One proton, one neutron and one electron

15. In an atom for the electron to revolve around the nucleus, the necessary centripetal force is obtained from the following force excerted by the nucleus on the electron:
(1) Nuclear force
(2) Gravitational force
(3) Magnetic force
(4) Electrostatic force

16. Ruther ford model could not explain the:
(1) Electronic structure of an atom
(2) Stability of an atom
(3) Both (1) and (2)
(4) None

17. On bombarding a beam of α particles on the atom of gold foil, a few particles get deflected whereas most of them go straight and remains undeflected. This is due to:
(1) The nucleus occupy much smaller volume as compared to the volume of atom
(2) The force of repulsion on fast moving α particles is very small
(3) The neutrons in the nucleus do not have any effect of α particles
(4) The force of attraction on α particles by the oppositely charged electron is not sufficient

18. According to the Thomson model of an atom, mass of the atom is assumed to be:
(1) Uniformly distributed over the atom
(2) Randomly distributed over the atom
(3) Partially distributed over the atom
(4) None of these

19. The spectrum of radiation emitted by a substance after absorbing energy is called:
(1) absorption spectrum
(2) emission spectrum
(3) white light spectrum
(4) none of the above

20. According to the Bohr's model of hydrogen atom:
(1) Total energy of electron is quantised
(2) Angular momentum of the electron is quantised and given as $\frac{nh}{2\pi}$
(3) Both (1) and (2)
(4) Neither (1) nor (2)

21. The frequency of EMW emitted by the revolving electron is equal to (According to classical theory):
(1) Double of the frequency of revolution
(2) Equal to the frequency of revolution
(3) Less than frequency of revolution
(4) Cannot say

22. If total energy of electron in an orbit is equal to E, is positive:
(1) Electron will revolve in an closed orbit
(2) Electron will not follow a closed orbit around the nucleus.
(3) Electron will be bound to nucleus
(4) Cannot say

23. Which one is not a failure of Bohr model:
(1) not applicable for more than one electron
(2) define frequency of radiation emitted by hydrogen atom
(3) Unable to explain the relative intensity of the frequency of spectrum
(4) Does not include electrical forces between electrons

24. Bohr's atomic model suggests that:
(1) Electrons have a particle as well as wave character
(2) Atomic spectrum of atom should contain only five lines
(3) Electron in hydrogen atom can have only certain values of angular momentum
(4) All of the above

25. According to Bohr theory the energy of an electron in the orbit:
(1) changes with time
(2) does not change with time
(3) change with pressure
(4) none of the above

26. The only series of lines appear in the visible region in the electro magnetic spectrum of hydrogen is:
(1) Lyman series (2) Balmer series
(3) Paschen series (4) Pfund series

27. The emission spectra of atoms in gaseous phase do not show a continous spread of wavelength from red to violet, rather they emit light only at specific wavelength with dark spaces between them. Such spectra is/are called:
(1) line spectra (2) atomic spectra
(3) both (1) and (2) (4) none of these

28. What kind of spectrum for electromagnetic radiation is obtained from solids and rarefied gases at a given temperature?
(1) Solids $\rightarrow$ Continuous ; gases $\rightarrow$ discrete
(2) Solids $\rightarrow$ discrete ; gases $\rightarrow$ discrete
(3) Solids $\rightarrow$ Continuous ; gases $\rightarrow$ continuous
(4) Solids $\rightarrow$ discrete ; gases $\rightarrow$ continuous

29. The most important contribution by Niel Bohr for explaination of Hydrogen atom was?
(1) Theory of stable orbits
(2) Angular momentum Quantization
(3) Energy level quantization
(4) e^- has dual nature

30. Choose the correct alternative :-

(1) Emission spectrum → White background with dark lines
Absorption spectrum → Dark background with bright lines

(2) Emission spectrum → Dark background with bright lines
Absorption spectrum → White background with dark lines.

(3) 1 & 2 both

(4) None of these

31. JJ Thomson's experiment revealed that

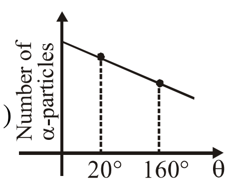
(1) atoms are smallest entities of matter

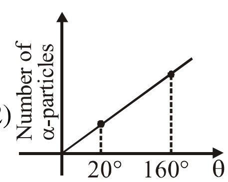
(2) atoms are round in shape

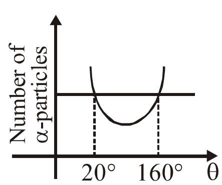
(3) atoms are not electrically neutral

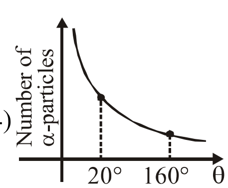
(4) atoms contains negatively charged particles

32. Graph of total number of α-particles scattered at different angles is

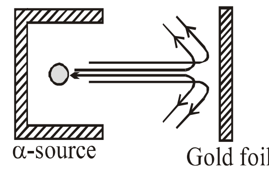
(1) 

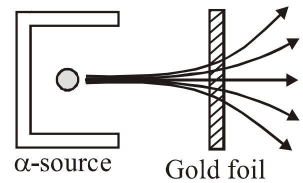
(2) 

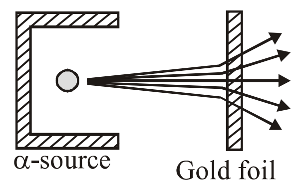
(3) 

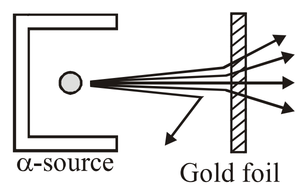
(4) 

33. Correct figure of Geiger-Marsden experiment is

(1) 

(2) 

(3) 

(4) 

34. Rutherford's atomic model could account for
I. concept of stationary orbits
II. the positively charged central core of an atom
III. origin of spectra
IV. stability of atoms
Choose the correct option from the codes given below.

(1) Only I is correct

(2) Only II is correct

(3) I, III and IV are correct

(4) All of these

35. A line emission spectrum of a material is

(1) unique like a fingerprint

(2) same for all

(3) may be same for few elements

(4) depends on excitation produced

36. According to classical electromagnetic theory emission spectrum of Rutherford atom must be

(1) a line emission spectrum

(2) a line absorption spectrum

(3) a continuous spectrum

(4) a band absorption spectrum

37. Which of these statements are correct regarding Bohr's model of hydrogen atom?

I. Orbiting speed of electron decreases as it shift to discrete orbits away from nucleus.

II. Radii of allowed orbits of electron are proportional to the principal quantum number.

III. Frequency of revolution of an electron is inversely proportional to the cube of principal quantum number.

IV. Binding force with which electron is bound to nucleus increases as electron shifts to outer orbits.

(1) I and III (2) II and IV

(3) I, II and III (4) II, III and IV

38. In Bohr's model where

(1) linear momentum is conserved while angular momentum is not conserved

(2) potential energy is conserved while kinetic energy is not conserved

(3) only potential energy is quantised

(4) only angular momentum is quantised

39. The de-Broglie wavelength of an electron in first Bohr's orbit is

(1) equal to $\frac{1}{4}$ of circumference of orbit

(2) equal to $\frac{1}{2}$ of circumference of orbit

(3) equal to twice of circumference of orbit

(4) equal to the circumference of orbit

40. The simple Bohr model cannot be directly applied to calculate the energy levels of an atom with many electrons as

(1) electrons not subjected to a central force

(2) electrons are colliding with each other

(3) electrons are subjected to screening effects

(4) force between nucleus and electrons is not subjected to Coulomb's force

41. In Bohr's atomic model, in going to a higher level (PE = potential energy, TE = total energy)

(1) PE decreases, TE increases

(2) PE increases, TE increases

(3) PE decreases, TE decreases

(4) PE increases, TE decreases

42. Bohr's second postulate defines the stable orbits on the basis of

(1) linear momentum of electron

(2) angular momentum of electron

(3) kinetic energy of electron

(4) total energy of electron

43. A set of atoms in an excited state decays

(1) in general to any of the states with lower energy

(2) into a lower state only when excited by an external electric field

(3) all together simultaneously into a lower state

(4) to emit photons only when they collide

44. When white light is passed through an unexcited gas (in rarified state), then transmitted light consists of

(1) few bright lines in dark background

(2) few dark lines in bright background

(3) alternate dark and bright lines

(4) alternate dark and bright bands

45. In an absorption spectrum, dark lines corresponds to same wavelengths which were found in emission line spectrum of gas. Above statement is

(1) correct for all gases

(2) correct only for hydrogen like atoms

(3) incorrect for all gases

(4) incorrect except for hydrogen